

Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

interaction_genotype_x_age.py

The mutation does nothing at 3 months and a great deal at 18 months, so the natural way to state that formally is an interaction: does the effect of the genotype depend on the age? The script fits a two by two model (mutant or wild-type, crossed with 3 months or 18 months) and then pulls out every useful comparison from that one model, so each gene can be classified by how its genotype effect changes as the fish get older.

- **interaction_genotype_x_age_results.csv:** the interaction result on its own, one row per gene with the interaction fold change and its q-value. The interaction fold change is a difference of differences: how much bigger the mutant versus wild-type gap is at 18 months than it was at 3 months. 4,395 genes come out significant.
- **interaction_gene_classification.csv:** the main table of the whole project. Every gene has the four group averages (wild-type at 3 months, mutant at 3 months, wild-type at 18 months, mutant at 18 months) with their error bars, so we can see whether the groups overlap at each age. Then it has five separate comparisons with fold changes and q-values: the genotype effect at 3 months, the genotype effect at 18 months, the age effect within wild-type, the age effect within mutant, and the interaction. Finally it has the labels, which say whether the gene is affected by age, by genotype, by both or by neither, and which pattern it follows. The headline is that 6,906 genes show age-dependent onset, meaning no difference at 3 months but a clear difference at 18 months, while zero genes are constitutive. Not one gene is affected by the mutation equally at both ages, so the entire genotype effect is something that emerges with age.
- **interaction_category_summary.csv:** just the counts, how many genes fall into each category, so the numbers can be quoted without recounting the big table each time.
- **interaction_lineplots.png:** the trajectory panels. Each panel is one gene, the x-axis has just two points (3 months and 18 months), the y-axis is normalised expression, and there are two lines, one for wild-type and one for mutant, with error bars. Lines that start together and then fan apart are exactly what an interaction looks like. Lines that stay parallel would mean the genotype effect is the same at both ages. tp53 is included as a panel because it was specifically asked about.
- **interaction_pattern_counts.png:** bar charts of how many genes fall into each category, both for the affected by labels and for the pattern labels. The bar that matters is the empty one, because constitutive comes out at zero.
- **interaction_geno_shift_scatter.png:** for every gene, the genotype effect at 3 months on the x-axis against the genotype effect at 18 months on the y-axis, coloured by pattern. Nearly all the coloured points pile up on the vertical line, meaning nothing at 3 months but a large effect at 18 months. The corners of the plot, which would hold genes affected the same way at both ages, are empty. It is the whole story of the project in one picture.